Day 23 | Friday

December 4, 2026

23

Angles, grades and carpentry slope applications

12:45 PM–2:15 PM • Math • 90 min

Week 6 • 1 scheduled block • 90 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:15 PM • 90 min

Angles, grades and carpentry slope applications

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–2:00 PM	**Practise**	35
2:00 PM–2:10 PM	**Check**	10
2:10 PM–2:15 PM	**Wrap up**	5

Math • Math Week 6, Lesson 9

One math block today, 12:45 PM–2:15 PM. Other scheduled program sessions remain separate.

Angles, grades and carpentry slope applications

Today’s goal: Find an angle from two sides and distinguish percentage grade from an angle in degrees.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Exercise Book 13A Q10–15 (p.128); 13B Q7–8 (p.129), Q12 (p.130). 13C Q4 (p.131), Q7 (p.132)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Recall and reconnect

Discuss whether a 2% slope is the same as 2°. Retrieve the opposite/adjacent ratio.

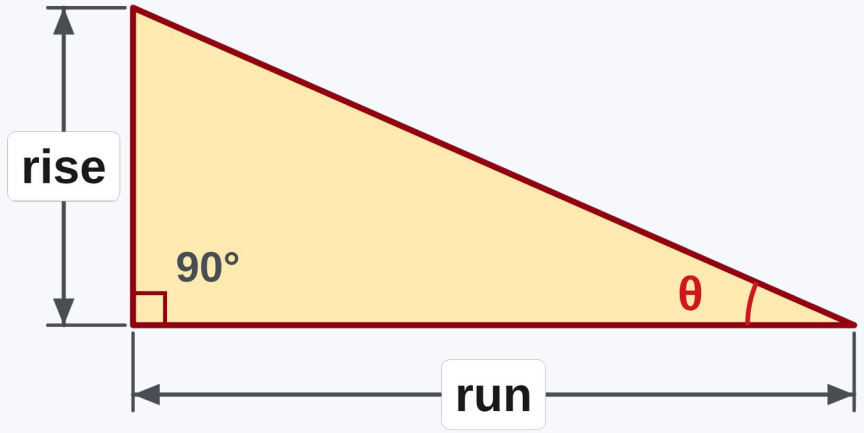
Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 13A Q10–15 (p.128); 13B Q7–8 (p.129), Q12 (p.130). 13C Q4 (p.131), Q7 (p.132)

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Find an angle from two sides

1. For rise and run, tan θ = rise/run.
2. Use inverse tangent to recover θ from the ratio.
3. Keep the calculator in DEG mode and label the angle with °.
4. Check that the calculated angle matches the relative rise and run.



$$\tan\theta = \text{rise/run}$$

$$\theta = \tan^{-1}(\text{rise/run})$$

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

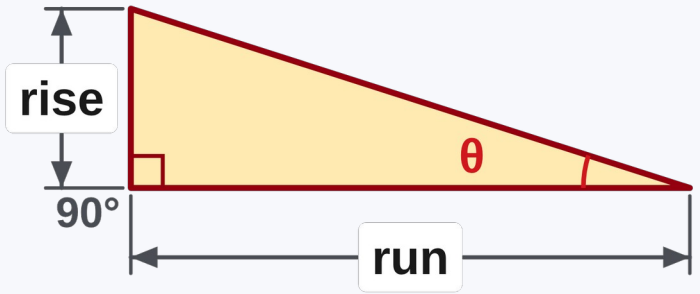
Keep grade and angle separate

1. Percentage grade = rise/run × 100.

2. Angle in degrees = $\tan^{-1}$(rise/run).

3. The same triangle gives two different descriptions of its slope.

4. Use consistent units for rise and run; label percentage and degree results separately.



Grade = rise/run × 100

report in %

Angle = $\tan^{-1}$(rise/run)

report in °

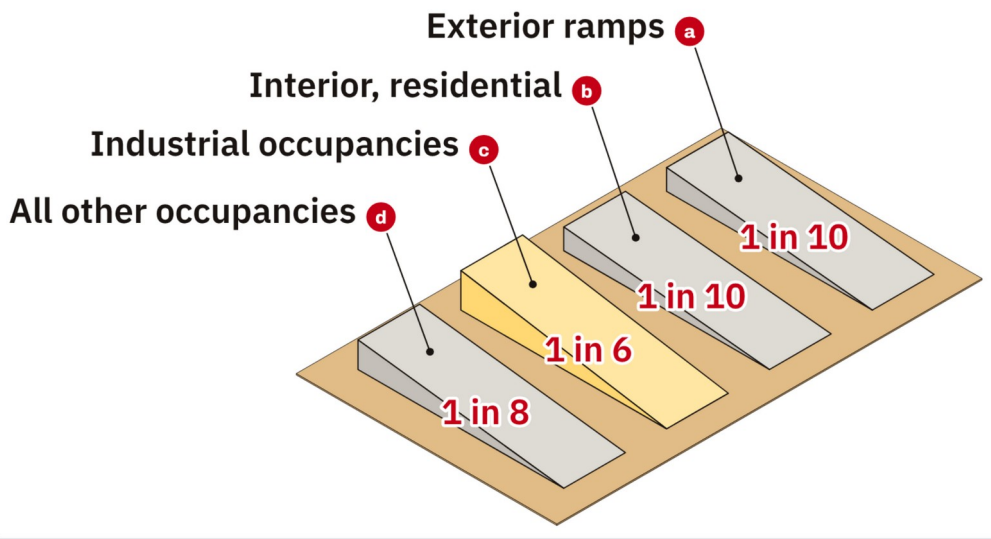
Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

One rise/run ratio, two different reports

Rise : run = 1 : 50 → rise ÷ run = 0.02
Grade = 0.02 × 100 = 2%
Angle = $\tan^{-1}(0.02) \approx 1.15°$

Angle example • Rise 350 mm; run 800 mm.
$\theta = \tan^{-1}(350 \div 800) \approx 23.63°$.
Use the inverse-tangent function, not 1 ÷ tan. Check DEG mode.

Together • Solve 13A Q10,14–15.
Label degree answers differently from percentage answers.
Shape Canvas: show rise and run; compare with a taught Numeracy Hub task.



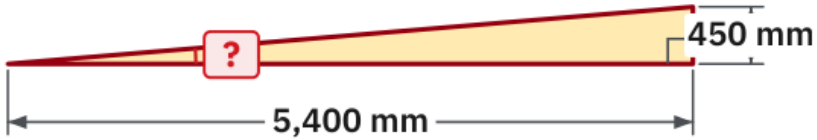
Drawn: four ramps with the same 3 m run, each at its steepest permitted slope.
Source: BCBC 2024, Division B, Article 9.8.5.4. (page 9-33)

Textbook pp.331–333 • Numeracy Hub → Angles and Trig → Right-triangle trigonometry

Textbook | page 333

Example 13.7 Angle and grade of a ramp

A ramp rises 450 mm over a run of 5,400 mm. Find its angle and its percent grade to two decimal places.



SOLUTION

1. Rise and run are opposite and adjacent: TOA.
2. Angle = $\tan^{-1}(450 \div 5{,}400)$.
3. Grade = $450 \div 5{,}400 \times 100 = 8.33\%$.

Textbook p.333 • Example 13.7 — Rise, run, angle and percentage grade

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Textbook | page 333

CALCULATOR

TI-30Xa

(450 ÷ 5400) 2nd TAN⁻¹

Display 4.763641691

Sharp EL-531XT

2ndF tan⁻¹ (450 ÷ 5400) =

Display 4.763641691

Answer Angle 4.76°, grade 8.33%. The drawing is to scale: a slope this gentle looks almost flat.

Textbook p.333 • Example 13.7 — Inverse tangent and separate degree/percent answers

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Spot the mistake. Explain the correction.

Does a 2% grade mean an angle of 2°?

Discuss the method before checking the answer.

No. Percent grade multiplies rise/run by 100; degrees come from inverse tangent in DEG mode.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Work together

Solve Exercise Book 13A Q10,14–15. Label degree answers differently from percentages.

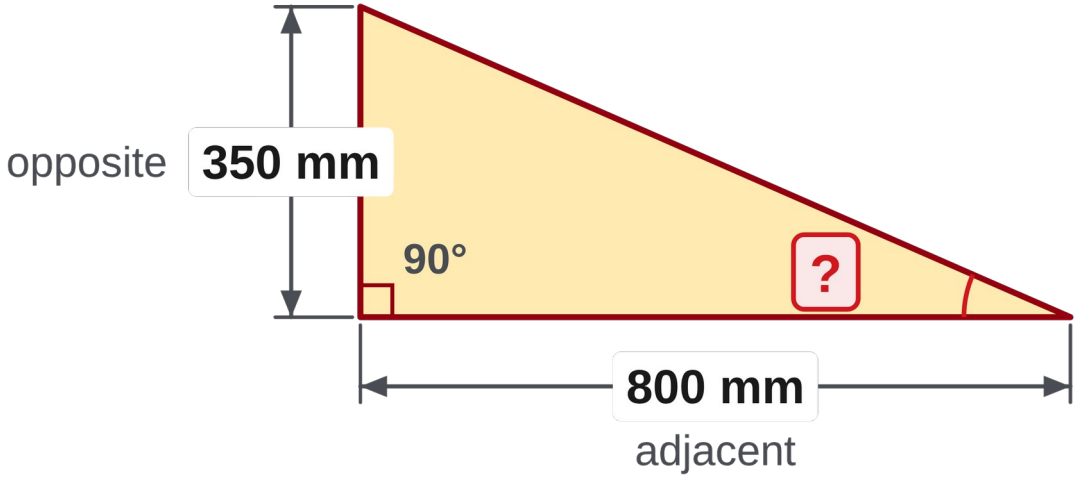
Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 13A Q10–15 (p.128); 13B Q7–8 (p.129), Q12 (p.130). 13C Q4 (p.131), Q7 (p.132)

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Guided practice | 13A Q10

Opposite 350 mm, adjacent 800 mm. Find the angle to two decimal places.



Work with a partner. Take turns explaining the next step.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Compare your method | 13A Q10

Opposite 350 mm, adjacent 800 mm. Find the angle to two decimal places.

CHECK TOGETHER

23.63°

$\tan^{-1}(350 \div 800)$

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Your scheduled practice

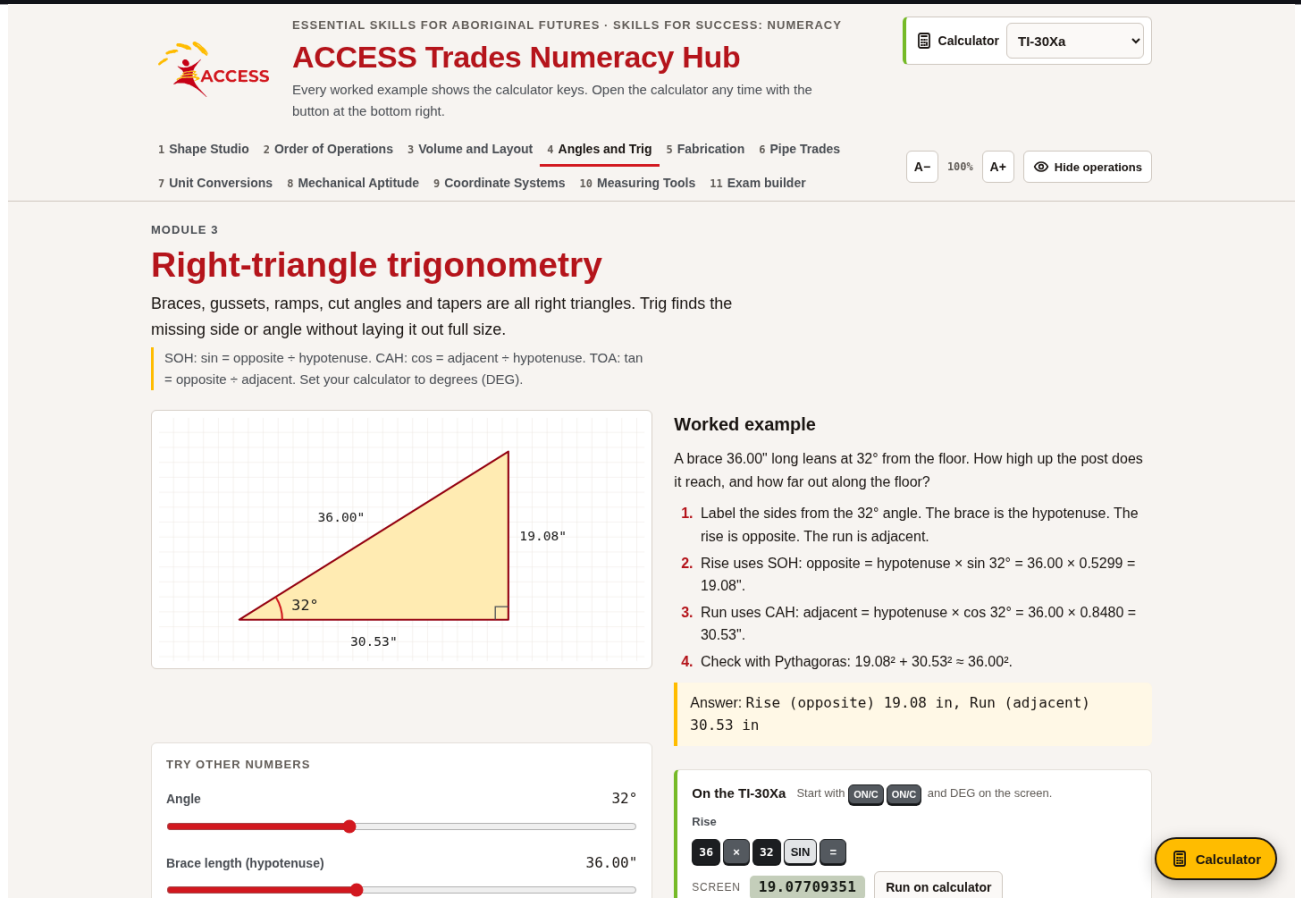
Complete Exercise Book 13A Q11–13 and 13B Q7; use Q12 as carpentry/truss extension. Show a Shape Canvas triangle and check the result with the Numeracy Hub. Then 13B Q8 and 13C Q4 and Q7: label each answer as a length, a percent grade or an angle in degrees.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 13A Q10–15 (p.128); 13B Q7–8 (p.129), Q12 (p.130). 13C Q4 (p.131), Q7 (p.132)

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Practise in Numeracy Hub and Shape Canvas

1. Build the assigned rise/run triangle in Shape Canvas and label both dimensions.

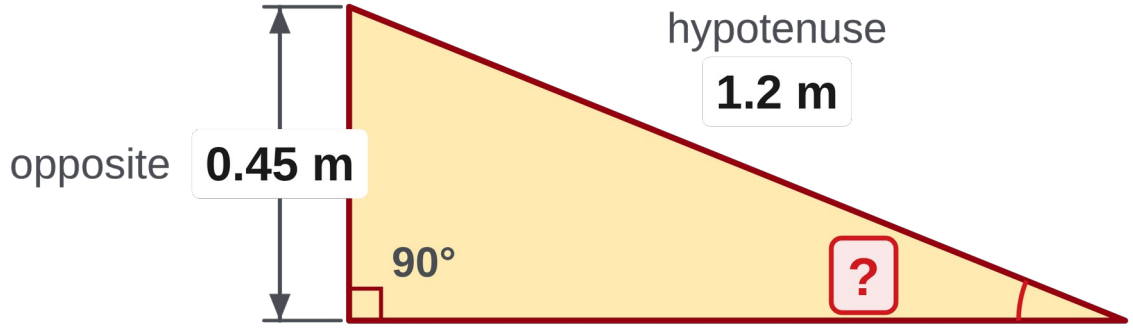
2. Calculate grade and angle separately on paper, keeping % distinct from °.

3. Check the triangle with Numeracy Hub Right-triangle trigonometry ramp/brace practice in DEG mode.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Practise from the Exercise Book | 13A Q11

Opposite 0.45 m, hypotenuse 1.2 m. Find the angle to two decimal places.

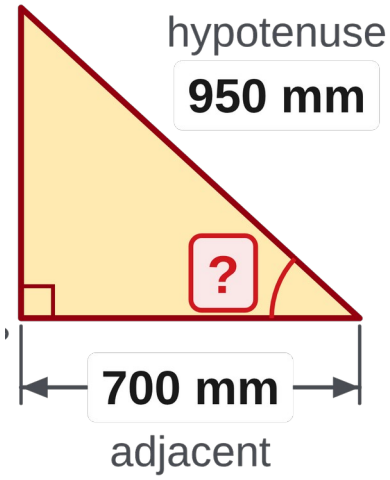


Write the operation, working and units before checking.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Practise from the Exercise Book | 13A Q12

Adjacent 700 mm, hypotenuse 950 mm. Find the angle to two decimal places.

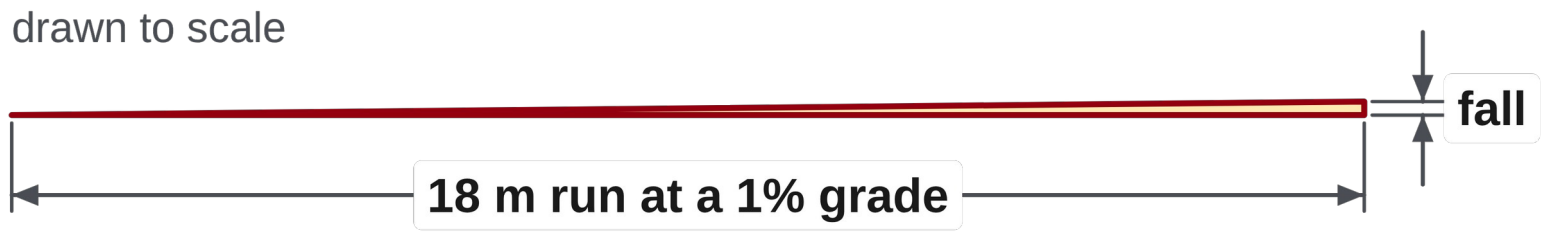


Write the operation, working and units before checking.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Practise from the Exercise Book | 13B Q8

A drain line is laid at a 1% grade over an 18 m run. How far does it fall, in millimetres?



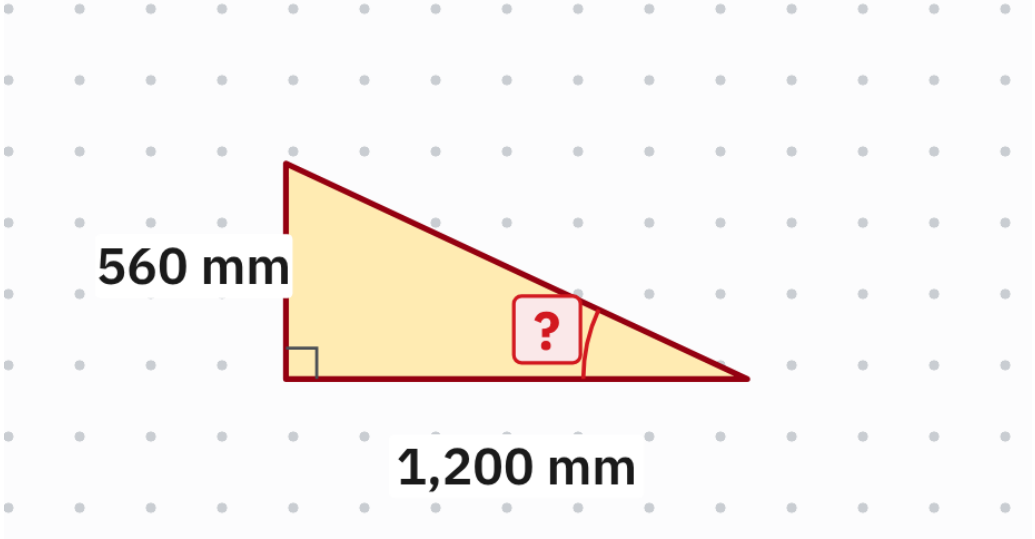
Fall in mm: ?

Write the operation, working and units before checking.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Practise from the Exercise Book | 13C Q4

Find the angle to two decimal places.

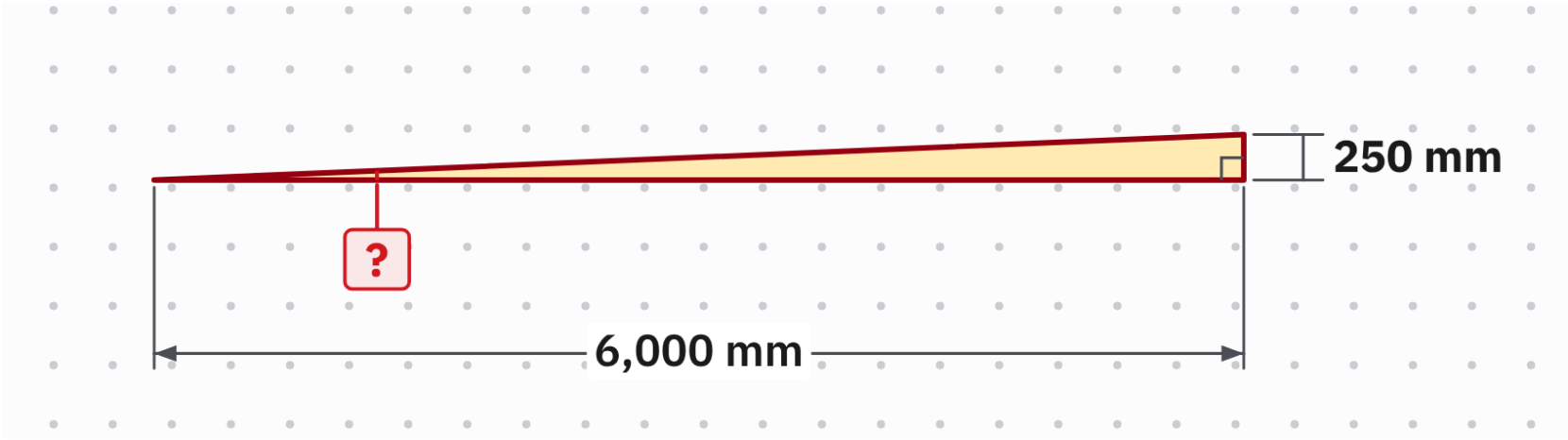


Write the operation, working and units before checking.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Practise from the Exercise Book | 13C Q7

A ramp rises 250 mm over a run of 6,000 mm. Find its angle to two decimal places. The drawing is to scale.



Write the operation, working and units before checking.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Check independently

Exercise Book 13A Q10 and Q14; give the grade and angle separately.

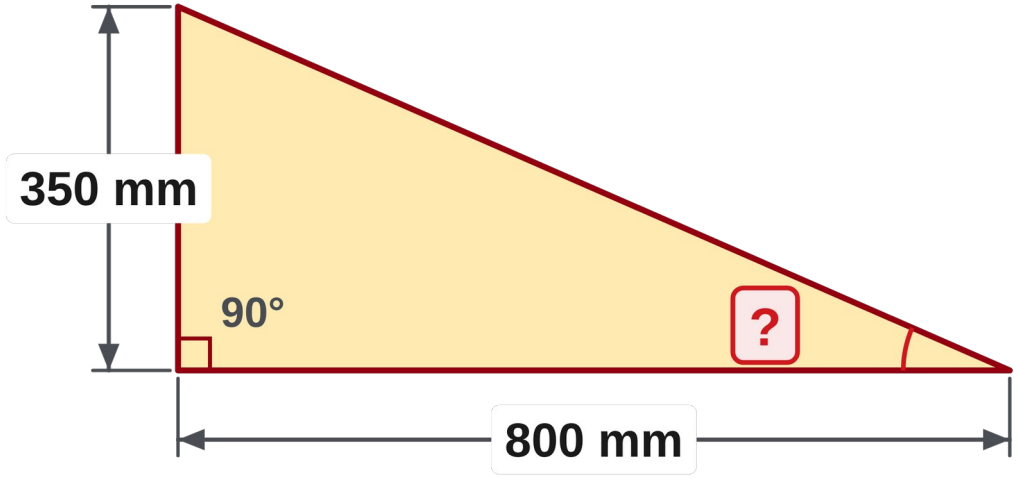
Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 13A Q10–15 (p.128); 13B Q7–8 (p.129), Q12 (p.130). 13C Q4 (p.131), Q7 (p.132)

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Independent check | 13A Q10

Opposite 350 mm, adjacent 800 mm. Find the angle to two decimal places.



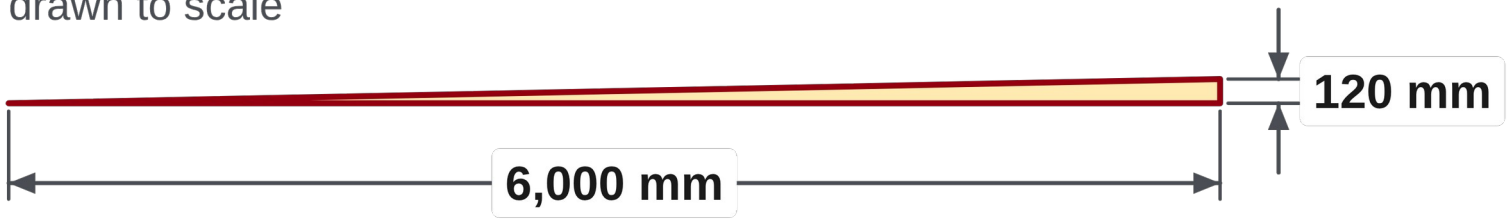
Answer independently. Keep worked solutions closed.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Independent check | 13A Q14

A slope rises 120 mm over a run of 6,000 mm. Find the percent grade and the angle to two decimal places.

drawn to scale



Percent grade: ?

Angle: ?

Answer independently. Keep worked solutions closed.

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 13A Q10–15 (p.128); 13B Q7–8 (p.129), Q12 (p.130). 13C Q4 (p.131), Q7 (p.132)

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)